

Screen Streaming & Seamless Takeover

Real-Time Browser Feeds and Operator Interventions for CAPTCHAs and 2FA

Governed Operator Collaboration: The Human Takeover Lease

When autonomous workers encounter high-security barriers—such as multi-factor authentication (2FA SMS), bank logins, or complex CAPTCHA puzzles—Fathom pauses gracefully and invites the human operator to assist.

1. LOW-LATENCY SCREEN FEED (/SCREEN)

The active browser viewport is streamed in real time over a WebSocket feed (500ms frame intervals) directly into the desktop app or web dashboard, allowing operators to monitor browser actions live.

2. EXCLUSIVE TAKEOVER LEASE (/CONTROL/WS)

Clicking **"Take Control"** grants the human operator exclusive keyboard and mouse access. While human-owned, bot automation commands are blocked, preventing accidental race conditions.

The 4-Step Collaborative Handshake

Phase 1

Agent Encounters 2FA Gate

The worker logs into a vendor billing portal; hits an SMS 2FA challenge. Emits a high-priority alert to the operator.

Phase 2

Operator Takes Control

The manager clicks "Take Control" in the desktop UI, enters the 6-digit SMS code from their phone, and completes the login.

Phase 3

Lease Released

The manager clicks "Release Control". The worker captures a fresh accessibility snapshot of the authenticated billing portal.

Phase 4

Autonomous Work Resumes

The digital employee continues downloading invoices and reconciling spreadsheets with zero loss of task context.

ZERO LOSS OF SESSION CONTEXT

Unlike traditional automation scripts that crash and require a complete restart when blocked, Fathom preserves accumulated session memory, active cookies, and reasoning variables during human intervention.

The Hybrid Ideal: 99% autonomous heavy lifting paired with instant 1% human oversight at critical security checkpoints.